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Affective Schemas: Acquisition, Updating, and Inference

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Schematized knowledge structures have been extensively studied in the cognitive domain, and yet the nature of affective schemas remains an uncharted area, with experimental work virtually nonexistent. Here, we examined how affective schemas are acquired, updated, and used for inference-making using three novel experimental paradigms. We show that affective schemas emerge by abstracting a common affective value from a distribution of unique affective associations. This common abstracted affective value semanticizes from the discrete exemplars into complex, valenced schemas (negative, positive, neutral), which consolidates across a 24-hr period. Valenced schemas (negative/positive) form faster than neutral schemas, resist affective reversals more strongly, and facilitate rapid learning and memory for related emotional information. Negative-valenced schemas, in particular, are most prioritized for learning, show greater resilience to change, and are more effective in supporting generalized (gist-based) inferences. This work defines key features of affective schemas, moving the study of emotional learning and memory systems from the conditioning of specific associations to the abstraction and consolidation of complex emotional knowledge.

Keywords: category learning, conditioning, emotion concepts, stress, semantic knowledge

Supplemental materials: https://doi.org/10.1037/emo0001561.supp

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Understanding how knowledge is learned, represented, and updated has been central to psychology for decades (Bartlett, 1932; Ghosh & Gilboa, 2014; Piaget, 1952; Schank & Abelson, 2013; Tulving, 1972). Cognitive science shows that semantic knowledge arises from the abstraction of commonalities across different experiences that are distilled into generalized concepts (Yee, 2019). For example, shared features of Poodles, Bulldogs, and Chihuahuas (four legs, fur, barks) are abstracted into the concept of a dog, ignoring breed-specific differences. Over time, a process known as *semanticization* links related abstracted concepts, forming coherent memory structures known as schemas (Gilboa & Marlatte, 2017). For example, schemas integrate physical concepts like "dog" with related concepts, such as "family pets" or "service animals." Schemas are not simply a collection of abstract concepts, but a rich associative network of interconnected meanings that guide human understanding, learning, and behavior across many contexts (Gilboa & Marlatte, 2017; Kumaran et al., 2009).

While cognitive schemas have been studied extensively (e.g., Audrain & McAndrews, 2022; Baldassano et al., 2018; Sommer, 2017; Sommer et al., 2022; van Kesteren et al., 2013), the features and mechanisms of affective schemas remain largely unknown, with experimental work virtually nonexistent. We distinguish affective schemas as memory structures where common internal affective, interoceptive, and motivational aspects of experiences *themselves*

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are semanticized into knowledge structures (Barrett, 2017; Fiske, 1982; Leahy, 2002; Tottenham, 2020; Vannucci et al., 2024). For instance, through semanticization, learning the specific associations of "dogs = happy feeling," "dogs = feeling of loyalty," and "dogs = emotionally supportive" connects related affective ideas evoking similar positive feelings, forming a generalized positive dog schema. Affective schemas, often nondeclarative and operating automatically (Tottenham, 2020; Vannucci et al., 2024), are essential for survival, affective meaning-making, and guiding behavior (Koban et al., 2021; Roy et al., 2012; Tottenham, 2020). For example, affective schemas may induce fear upon encountering a never-before-seen animal because it shares features with a snake. Uncovering the basic mechanisms of affective schemas can advance foundational principles of human cognition and behavior. Here, we sought to characterize the properties of affective schema acquisition, updating (assimilation and accommodation), and inference across three novel experimental paradigms (Figure 1).

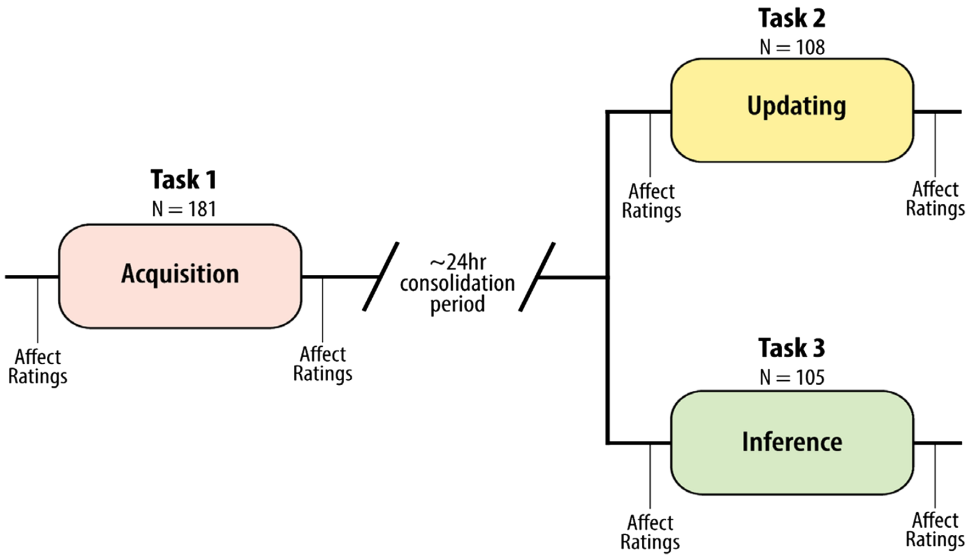
Forming Affective Schemas

Existing work on schema acquisition in humans has focused primarily on two types of cognitive tasks: (a) spatial arrays of object–location paired associates (Sommer, 2017; Sommer et al., 2022) and (b) rule-based relationships using tasks such as weather prediction (Wagner et al., 2015), hierarchies between items (Brod et al., 2015, 2017), and sequential character presentation (Pudhiyidath et al., 2020).These tasks are affectively neutral. Similar tasks that do employ affect tend to rely on category conditioning tasks, which typically associate known concepts (animals/tools) with an aversive shock (Boyle et al., 2016; Dunsmoor et al., 2012; Patterson et al., 2022) or monetary reward (Oyarzún et al., 2016; Patil et al., 2017). This line of work shows that conceptual categories can be associated with a single affective stimulus. In contrast, affective schemas are constructed from *multiple unique* affective stimuli (e.g., aversive shock, angry yelling, sad crying) with a common underlying affective value (e.g., negative valence) that can be associated with other concepts (e.g., animal subspecies). Task 1 accomplishes this by modifying a traditional classical conditioning (Andreatta & Pauli, 2015; Bach et al., 2023) paradigm to compare the dynamics of acquiring positive, negative, and neutral schemas over a 24-hr consolidation period. A 24-hr consolidation period was chosen based on evidence that such delays preferentially enhance the consolidation of affective memory traces (Payne & Kensinger, 2018; Yeh et al., 2021). Memories also undergo a dynamic transformation within this window, shifting from episodic-like subcortical representations to more abstract, generalized cortical representations that support stable schematized knowledge structures (Robin & Moscovitch, 2017).

Updating Affective Schemas

An essential feature of schemas is their adaptability (Ghosh & Gilboa, 2014), so Task 2 investigated how affective schemas are flexibly updated through processes of assimilation and accommodation. Cognitive schemas improve learning of schema-consistent and, at times, schema-inconsistent information compared to irrelevant information (Gisquet-Verrier & Riccio, 2018; McClelland et al., 2020;

Figure 1
Study Overview



Note. Experimental Tasks 1–3 involved distinct, yet overlapping samples (for detailed participant flow, see the procedures section and Supplemental Figure S1). During Visit 1 participants completed Task 1, an affective schema acquisition task. Visit 2 occurred ~24 hr after initial acquisition to allow time for consolidation and abstraction. At Visit 2, participants completed Task 2, an affective schema updating task, and/or Task 3, an affective schema-based associative inference task. For participants who completed both Visit 2 tasks, the inference task was completed before the updating task so that the initial schema acquired at Visit 1 was intact. Participants completed affective valence ratings of 12 schema exemplars before and after each task to measure within-subject changes in the affective components of the schema representations. See the online article for the color version of this figure.

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Yaple & Yu, 2019). Schema assimilation integrates new, schema-consistent information without significant changes to the existing knowledge structure (Piaget, 1952; Sommer et al., 2022). For example, a child with a positive pet schema easily incorporates a new experience with a friendly cat. When salient new information is schema-inconsistent, schema accommodation must occur to learn the new information (Piaget, 1952; van Kesteren et al., 2013); if the cat is aloof and scratches the child, the affective schema is accommodated to understand that cats *can* be associated with positive-valence feelings but they *can also* be associated with negative-valence feelings. Accommodation of affective schemas likely share principles with conditioning processes that involve replacing an expected unconditioned stimulus with one of a different valence (e.g., counterconditioning, Keller et al., 2020, reversal learning, Yaple & Yu, 2019, novelty-facilitated extinction, Lipp et al., 2020). We integrated these conditioning designs with schema learning tasks to characterize affective schema updating in Task 2. Alongside old stimuli, we introduced new affective schema-consistent and -inconsistent stimuli to probe assimilation and accommodation, respectively.

Inference From Affective Schemas

A hallmark of schemas is their ability to facilitate novel inferences to learn about new situations, a process supported by memory integration. Memory integration forms overlapping associations between new information and related preexisting representations, allowing for novel connections to be made within a schema's flexible structure (McClelland et al., 2020; van Kesteren et al., 2020). Associative inference tasks, which involve encoding overlapping item pairs (A–B and B–C) and then inferring A–C associations, are a gold standard for studying episodic memory integration (Preston et al., 2004; Schlichting et al., 2017, 2022; Spalding et al., 2018). The role of affective schema knowledge in associative inference has yet to be examined. The impact of emotional information on memory integration and inference-making is also unclear (de Vries et al., 2022; Shohamy & Daw, 2015; Zhu et al., 2022), though it likely plays a crucial role in affective meaning-making. For instance, a positive dog schema semanticized from pleasant experiences might help infer that interacting with the dog's owner is also enjoyable. Task 3 adapted the classic associative inference paradigm (Schlichting et al., 2017) to examine how affective schemas (negative and positive vs. neutral schemas) support inference-making. Here, inferences were tested by examining whether the original A–B pairs that created the affective schemas, together with newly encoded B–C pairs, would facilitate novel A–C affective inferences.

The design of Task 3 also enabled us to investigate whether gist-level memories related to affective schemas alone were sufficient to make novel affective inferences. In the cognitive domain, gist traces capture fuzzy conceptual content (such as the essential meaning, structure, or "crux" of an experience) and facilitate memory for schema-consistent information (Reyna & Brainerd, 1998; Webb et al., 2016). At the neural level, these coarse "global" gist memories may be encoded alongside richly detailed "local" episodic memories, enabling flexible memory integration and retrieval through dynamic interactions between schematic and episodic memory systems (Moscovitch & Gilboa, 2024; Sekeres et al., 2018). In the emotional domain, affective information has been shown to promote semantic false memory errors through gist-based inferences (Mirandola & Toffalini, 2016; Vannucci et al., 2024), perhaps because emotional experiences prioritize the broader meaning of events and lead to greater reliance on gist representations (Bookbinder & Brainerd, 2016, 2017). Given that affective schema memories are, by definition, generalized representations, we hypothesized that gist-level information alone suffices for novel affective inferences when access to detailed memories is unavailable. To test this hypothesis, Task 3 adapted the associative inference task to assess A–C inference performance both when participants showed intact B–C memories for detailed information and when they only retained gist-level, but not verbatim, B–C memories.

The Current Research

The current research investigates the properties of affective schemas. We demonstrate the nature of affective schema acquisition, affective schema updating, and the role of affective schemas in inference-making across three preregistered experiments. Our novel tasks leverage established conditioning (Bach et al., 2023; Dunsmoor et al., 2012; Keller et al., 2020; Yaple & Yu, 2019) and schema (Brod et al., 2015, 2017; Schlichting et al., 2017) paradigms with known effects to maximize validity and allow comparisons to prior work. *Task 1* examines how negative, positive, and neutral schemas are initially acquired. Participants learned to associate "Fribble" families (Barry et al., 2014) with classes of emotionally evocative unconditioned stimuli (i.e., noises from a standardized affective sound database; Yang et al., 2018). Each family was paired with classes of either negative, positive, or neutral noises. Negative and positive noises were matched on arousal (and higher than neutral noises), and since arousal and valence facilitate episodic-like associative learning and memory (Clewett & Murty, 2019; Kensinger & Corkin, 2004), we expected faster acquisition of valenced affective schemas compared to neutral ones but relatively few differences between negative and positive schemas.

Tasks 2 and 3 investigate how the consolidation of affective schemas (after ∼24 hr; Brod et al., 2017; Kensinger & Ford, 2020) influences new learning. A consolidation period of 24 hr was selected in light of literature showing that such delays facilitate the preferential consolidation of affective memory traces. *Task 2* investigates updating of affective schemas through assimilation and accommodation of new schema-consistent or -inconsistent stimuli. As observed in the cognitive domain (Sommer et al., 2022; van Kesteren et al., 2013), we expected affective schema-consistent new stimuli to be learned as well as old stimuli and better than schema-inconsistent stimuli. Accommodation was tested with valence reversals (negative-to-positive and positive-to-negative), neutral-to-valenced reversals (neutral-to-negative/positive), and valenced-to-neutral reversals (negative/positive-to-neutral). Since negative- and positive-valence concepts are drastically distinct from one another (Russell, 1980), we expected valence reversals to be the most resistant to accommodation. *Task 3* adapts the associative inference paradigm to examine how affective schemas support new learning and affective inferences, testing whether gist representations are sufficient when verbatim memories are inaccessible. We hypothesized that gist-level affective inferences are more likely for negative and positive schemas than for the neutral one. Through these experiments, we elucidate how affective schemas form, adapt, and operate, advancing our understanding of complex emotional memory.

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Method

Participants

The sample comprised 181 participants (age: M = 23.79 years, SD = 4.97, range = 18–36; 54.7% female, 43.1% male, 2.2% other gender identity) who completed the study either in the laboratory or online (see Supplemental Table S1 for sample descriptives and Supplemental Figure S2 for participant flow). The race/ethnicity breakdown was 19.9% (n = 36) Asian or Asian American, 11.1% (n = 14) Black or African American, 13.5% (n = 17) Hispanic/Latine, 44.4% (n = 94) non-Hispanic White or Caucasian, 10.5% (n = 19) multiracial/ethnic, and 0.6% (n = 1) other/unknown.

Laboratory Sample

A total of 55 undergraduates (age: M = 21.76 years, SD = 4.57, range = 18–36) were recruited from introductory psychology classes at a large urban university in the northeast region of the United States. There were no exclusion criteria.

Online Sample

A total of 126 young adults from the United States (age: M = 24.70 years, SD = 4.88, range = 18–35) were recruited via Prolific (https://www.prolific.co). Participants were eligible if they were between the ages of 18–36 years (to match the age range of the laboratory sample), resided in the United States, and willing to participate in a 2-day study. Participants were excluded if they did not have a laptop or headphones to complete the study procedures.

Procedure

This study included data from two samples and took place over two visits on consecutive days (Figure 1). At Visit 1, participants completed questionnaires via Qualtrics (Qualtrics, Provo, Utah; https://www.qualtrics.com) and the affective schema acquisition task. The next day, at Visit 2, participants completed the inference and/or updating tasks. Participant retention from Visit 1 to Visit 2 was high (90.1%). Visit 2 occurred, on average, 23:35 hr after Visit 1, with no differences between experiment modality (laboratory: 23:37 vs. online: 23:34). All participants provided written informed consent. The affective schema tasks were coded in python with the Psychopy Builder v3.1.5 (Peirce et al., 2019) and hosted via Pavlovia (*Pavlovia*, 2020). The study was performed in accordance with the Declaration of Helsinki and approved by the Columbia University' institutional review board.

Laboratory Sample

Students from introductory psychology courses were invited to participate in the study through the psychology department's Participation Pool via Sona Systems (https://www.sona-systems.com/). Participants signed up for hour-long time slots on two consecutive days and received course credit upon study completion. Participants completed all three affective schema tasks (acquisition, updating, and inference). All procedures took place in a sound-proofed room in an on-campus laboratory between January 2020 and March 2020. The in-person study was halted on March 7, 2020 due to a university-wide shutdown of in-person research in response to the COVID-19 pandemic.

Online Sample

Data collection occurred in December 2020 and participants received $25 upon study completion ($10 for Visit 1; $15 for Visit 2). For Visit 2, participants were randomly assigned to complete either the updating task (n = 58) or the inference task (n = 58) to minimize fatigue/effort concerns (each task took ~20–25 min). The following evidence-based strategies (Kominsky et al., 2021) were taken to mitigate the potential pitfalls of online data collection: (a) the task could only be completed on a laptop/desktop computer; (b) interactive visual instructions; (c) practice trials examined participants' understanding of the task; and (d) all responses and reaction times were recorded. With these precautions taken (Kominsky et al., 2021), unmoderated online experiments have replicated in-lab findings from affective, learning, and memory tasks (Giraudier et al., 2022; Nussenbaum et al., 2020; Vales et al., 2021).

Experimental Stimuli

The affective semantic knowledge, operationalized here as affective schemas, was that different animallike species (Fribbles) had a negative, positive, or neutral valence (i.e., "good," "bad," or "neutral"), which was determined by their physical attributes and noises they made.

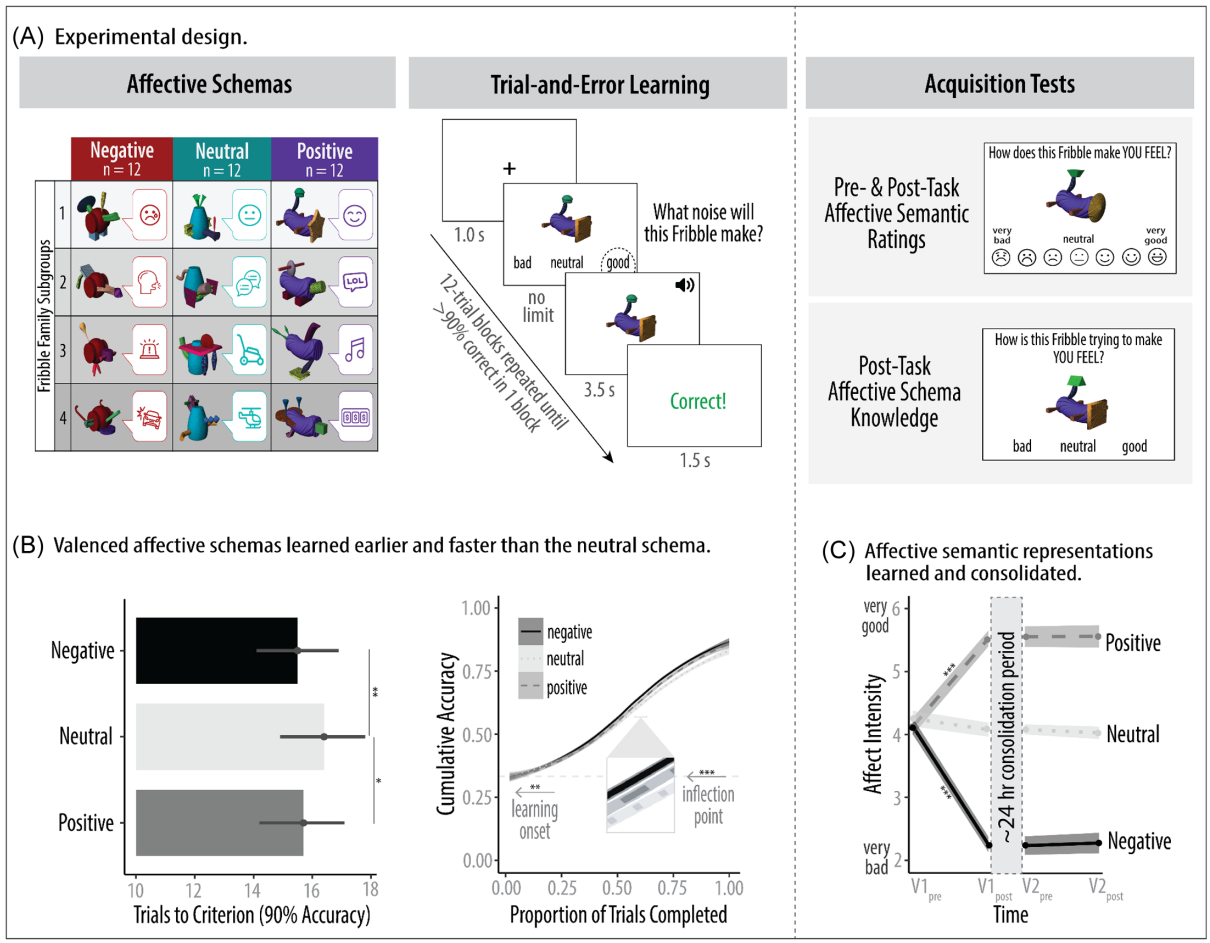
Fribbles

Fribbles were three-dimensional combinations of different colors, shapes, and textures that resemble real-world animals (Barry et al., 2014). Fribbles were designed with a species structure based on the color of their bodies and differing appendages (12 species in total; Figure 2A, left). A total of 973 unique Fribbles stemmed from 81 different exemplars of each species because all appendages have three different variations (e.g., head shape of a circle, square, and triangle). Fribbles were designed for use in visual experiments and have been used in schema learning studies with children and adults (Brod et al., 2015, 2017). Fribbles were optimal for schema learning experiments because they were similar to real-world stimuli but participants did not have prior experience with them. In the current experiment, participants were not exposed to the same Fribble more than once to ensure generalized/semantic learning and memory processes, with the exception of the 12 Fribbles used for Fribble affect semantic ratings and the Visit 1 Fribbles used in the updating task to test assimilation hypotheses.

Noises

For the affective semantic component of the schemas, a range of emotionally evocative auditory stimuli were selected from the expanded International Affective Digitized Sounds stimulus database (Yang et al., 2018). The noises in this experiment comprised six broadly unpleasant "bad" noises (e.g., crying, car wreck), six pleasant "good" noises (e.g., giggle, music), and six "neutral" noises (e.g., yawn, typewriter; Supplemental Table S2). The noises were selected based on valence and arousal ratings from pilot testing with 20 adults. The unpleasant and pleasant noises were matched on

Figure 2
Affective Schema Acquisition



Note. Results of Bayesian multilevel models for the affective schema acquisition task. Points are posterior means, while ribbons and error bars are 95% credibility intervals. In Bayesian models, the strength of the evidence for the alternative hypothesis (H1) over the null hypothesis (H0) is indicated by probability of direction estimates (p_{dir} values). A. *Left*: The affective schema involved learning that some Fribble families made negative, positive, or neutral noises based on the color and shape of their body. *Middle*: Participants completed a trial-and-error affective schema learning task with feedback until reaching the schema acquisition criterion (90% accuracy in a block of 12 trials). *Right:* Affective schema knowledge was tested with (a) affective ratings of how each Fribble made participants feel in the moment before and after the acquisition phase and (b) a valence knowledge test that had the same structure as the acquisition task except that no feedback was provided. B. *Left*: Participants reached the acquisition criterion for negative and positive schemas in fewer trials than for the neutral schema. *Right*: Group-level estimate of sigmoid curves using trial-by-trial cumulative accuracy scores. Learning onset (horizontal shift parameter) occurred earlier for the positive (vs. neutral, negative) schema, whereas the negative schema was associated with an earlier inflection point (see Supplemental Figure S3C for parameter plots). C. Participants reported affective changes in expected, schema-consistent directions from pre- to postacquisition task; these ratings were maintained after ~24 hr and persisted following the associative inference task (during which no changes in affective semantic content were expected). vs. = versus. Icons licensed from Freepik (https://www.freepik.com/). See the online article for the color version of this figure.
* p_{dir} = .95–.96. ** p_{dir} = .97–.98. *** p_{dir} = .99–1.00.

arousal level. To minimize confounds related to sound type, each condition (negative, positive, and neutral) included equal numbers of human vocalizations and nonhuman object-generated sounds (Plichta et al., 2011). Balancing the diversity of affective stimuli across conditions should reduce the likelihood that within-schema relatedness accounts for observed effects (Talmi, 2013). The use of multiple, diverse affective stimuli distinguishes the current schema learning paradigm from prior classical and category conditioning tasks (Dunsmoor et al., 2012; Klein et al., 2022), which utilize one specific aversive or appetitive stimulus during the acquisition phase (e.g., mild shock, aversive noise). In other words, affective knowledge *itself* is generalized in nature and spans across three abstract, emotional valence concepts (negative, positive, and neutral/nonvalenced).

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Visit 1: Forming Affective Schemas

Experimental Design

The affective schema acquisition task was based on merging the experimental designs of classical and category conditioning tasks (Dunsmoor et al., 2012; Klein et al., 2022) with a nonaffective schema learning paradigm (Brod et al., 2015, 2017). Participants learned the affective schema that Fribbles were "bad" (made unpleasant noises), "good" (made pleasant noises), or "neutral" (made neutral noises) based on shared physical characteristics. The body color-valence pairings for the affective schema were randomly counterbalanced across participants. The 36 Fribble exemplars that looked most different from each other were used for the affective schema acquisition task (Barry et al., 2014). Participants were instructed to find out which Fribbles were (a) nice and made good noises, (b) mean and made bad noises, and (c) calm and made neutral noises via an interleaved, trial-and-error feedback learning task (Figure 2A, left). Fribbles were presented one at a time (until the participant made a response), in random order, in three counterbalanced blocks of 12 (four bad Fribbles, four good Fribbles, and four neutral Fribbles per block, for a total of 36 novel Fribbles). When a Fribble appeared on the screen, participants were asked to predict what type of noise they expected the Fribble to make: "bad," "good," or "neutral" (Figure 2A, middle). This was followed by the Fribble immediately eliciting the correct noise for 3.5 s and explicit feedback was displayed for 1.5 s ("Correct!" for accurate predictions vs. "Oops! You made a mistake." for inaccurate predictions), which was followed by a 1 s fixation cross (Figure 2A, middle). Participants were required to complete the three blocks once. Beyond this minimum, blocks were repeated until the participant reached a block performance of 90% accuracy. Reaching the 90% block-level accuracy criterion was an a priori indicator that participants learned the affective schemas.

Affective Schema Knowledge Test

After reaching the acquisition criterion, participants completed a three-alternative, forced-choice affective schema knowledge test (Figure 2A, right). They were presented with 36 novel Fribbles one at a time and were asked to select how each Fribble was trying to make them feel (good, bad, or neutral). No noises or corrective feedback were provided. Response accuracy was interpreted as explicit knowledge of the affective schemas. Correct responses also indicated generalization of the affective schemas to new, perceptually similar Fribbles.

Fribble Affective Semantic Ratings

To evaluate the extent to which the affective semantic components of the schemas were acquired and generalized to new stimuli, participants rated how 12 different Fribbles (one novel exemplar from each of the 12 species) made them feel *in the moment* both before and after the acquisition task (Figure 2A, right). Affective intensity ratings were made on a 7-point Likert-type scale, ranging from 1 = *very bad* to 7 = *very good*. The Fribble affect ratings were averaged as a function of affective schema valence (positive, negative, and neutral) at each timepoint. The outcome was the pre- to posttask change in ratings.

Visit 2: Updating Affective Schemas

Experimental Design

Affective schema updating processes (assimilation, and accommodation) were investigated using the same trial-and-error feedback design as the acquisition task (Figure 3A). Participants were told that they would see some of the same Fribbles they had previously encountered (*old*), as well as other new Fribbles from a related family that share physical similarities with the old Fribbles (*new*). A total of 240 trials were presented in five blocks of 48 trials that were randomly counterbalanced across participants.

There emotional schema updating conditions were: (a) *old-consistent*: 60 original Fribbles repeated from Day 1 that made noises consistent with the affective schema (20 negative, 20 positive, 20 neutral); (b) *new-consistent:* 90 new Fribbles that made noises consistent with the affective schema (30 negative, 30 positive, 30 neutral); and (c) *new-inconsistent*: 90 new Fribbles that made noises contradictory to the affective schema. The new-inconsistent Fribbles changed their emotional valence as a function of their subspecies that had distinct appendages (Figure 3C). Among the new-inconsistent Fribbles, there were six affective schema adaptation conditions each comprised of 15 different Fribbles: (a) positive-to-neutral (Pos → Neu), (b) negative-to-neutral (Neg → Neu), (c) neutral-to-positive (Neu → Pos), (d) neutral-to-negative (Neu → Neg), (e) positive-to-negative (Pos → Neg), and (f) negative-to-positive (Neg → Pos).

As with the acquisition task completed at Visit 1, participants were instructed to predict which Fribbles were (a) nice and made good noises, (b) mean and made bad noises, and (c) calm and made neutral noises. For each trial, participants made their prediction (good, bad, or neutral), the noise played for 3.5 s, explicit corrective feedback was provided for 1.5 s, and a fixation cross appeared for 1 s (Figure 3A).

Fribble Affect Semantic Ratings

Both before and after the updating task, participants completed Fribble affect ratings using the same Visit 1 procedures to assess the extent to which the affective semantic components of the schemas were updated.

Visit 2: Inferences From Affective Schemas

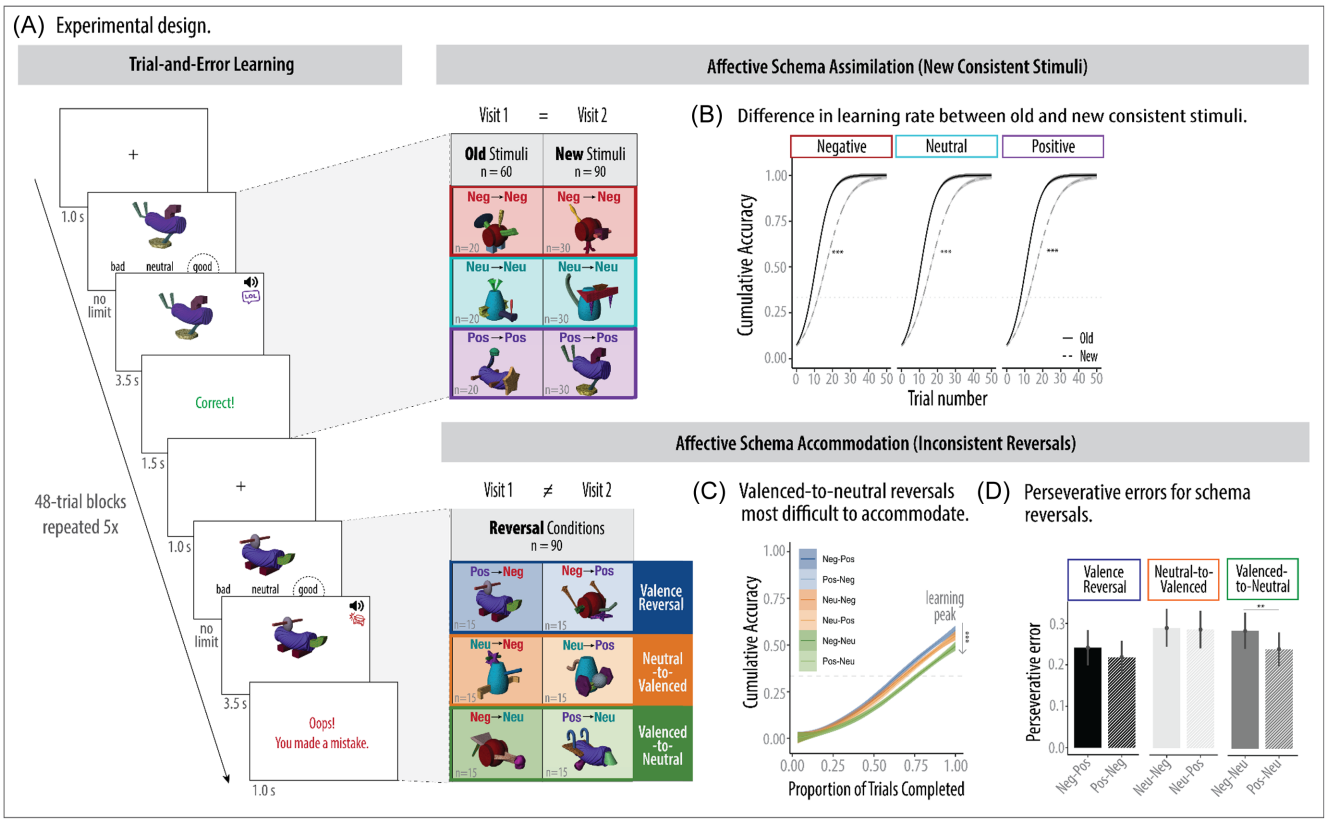
Experimental Design

The stimuli were affective noises (A), new Fribbles (B), and black shapes (C) that were arranged into ABC triads. The ABC triplets comprised two directly studied, overlapping pairs (A–B and B–C) as well as nonstudied associative inference pairs (A–C; Figure 4A). The Fribble-shape (B–C) pairs and stimulus presentation were randomized. Procedures were adapted from associative inference tasks used in prior cognitive neuroscience studies (Schlichting et al., 2022; Spalding et al., 2018). The associative inference task occurred in five stages: A–B study phase, B–C study phase, A–C test phase, B–C test phase, and A–B test phase.

A–B Study Phase (Affective Schemas)

The affective schemas, comprising noise + Fribble pairs, were learned and tested at Visit 1 (see above). At the beginning of Visit 2,

Figure 3
Affective Schema Updating



Note. Results of Bayesian multilevel models for the affective schema updating task. Points are posterior means. Ribbons and error bars are 95% credibility intervals. The strength of the evidence for the alternative hypothesis (H1) over the null hypothesis (H0) is indicated by probability of direction estimates (p_{dir} values). A. Affective schema updating processes (assimilation and accommodation) were examined with the same interleaved, trial-and-error feedback design as the acquisition task. Some Fribbles were previously encountered (i.e., "old"), but most were new Fribbles from the same families who made noises that were either *consistent* or *inconsistent* with the prior affective schema knowledge established during the Visit 1 acquisition task. B. Old and new schema-consistent stimuli were compared to examine affective schema assimilation. The learning peak was near-perfect across all affective schemas, suggesting robust assimilation of new, similar experiences into prior affective knowledge. Yet, novel schema-consistent stimuli were learned slower than old, previously encountered stimuli. C. For schema-inconsistent stimuli, the valence of some Fribble subfamilies (i.e., valence of the noises) within each affective schema were reversed to examine accommodation, which included valenced reversals (negative-to-positive and positive-to-negative), neutral-to-valenced reversals (neutral-to-negative and neutral-to-positive), and valenced-to-neutral reversals (negative-to-neutral and positive-to-neutral). Results indicated that valenced-to-neutral schema reversals were associated with lower accuracy, suggesting that more difficulty with accommodation. By contrast, valenced schema reversals were associated with the highest accuracy. D. Perseverative errors (i.e., incorrect responses indicating the valence of the existing schema) probed affective schema resistance to accommodation. Perseverative errors were more frequent for valenced-to-neutral reversals when the original affective schema was negative as opposed to positive. Icons licensed from Freepik (https://www.freepik.com/). See the online article for the color version of this figure.
** p_{dir} = .97–.98. *** p_{dir} = .99–1.00.

participants completed Fribble affect ratings using the same Visit 1 procedures to assess the affective schema. Subsequently, the affective schema acquisition procedures (see above) were repeated with new Fribbles to ensure the instantiation of the affective schema knowledge.

B–C Study Phase (Pair Memory)

Participants were instructed to learn that 36 unique B–C pairs of new Fribbles and black shapes belonged together. Participants were encouraged to create narratives or use visual imagery to help them remember the Fribble-shape pairs, but they were not required to respond explicitly. Each Fribble-shape (B–C) pair was presented for 12 s followed by a 1.5 s fixation cross, which was repeated twice.

A–C Inference Test

The emotional valence (Item A) and shapes (Item C) were linked implicitly through their common association with the Fribble species (Item B), but they never appeared together. This associative structure resulted in a hidden relationship that required participants to make inferences based on the affective component of the schemas. Participants completed a three-alternative, forced-choice inference task in which they were instructed to select the emotional

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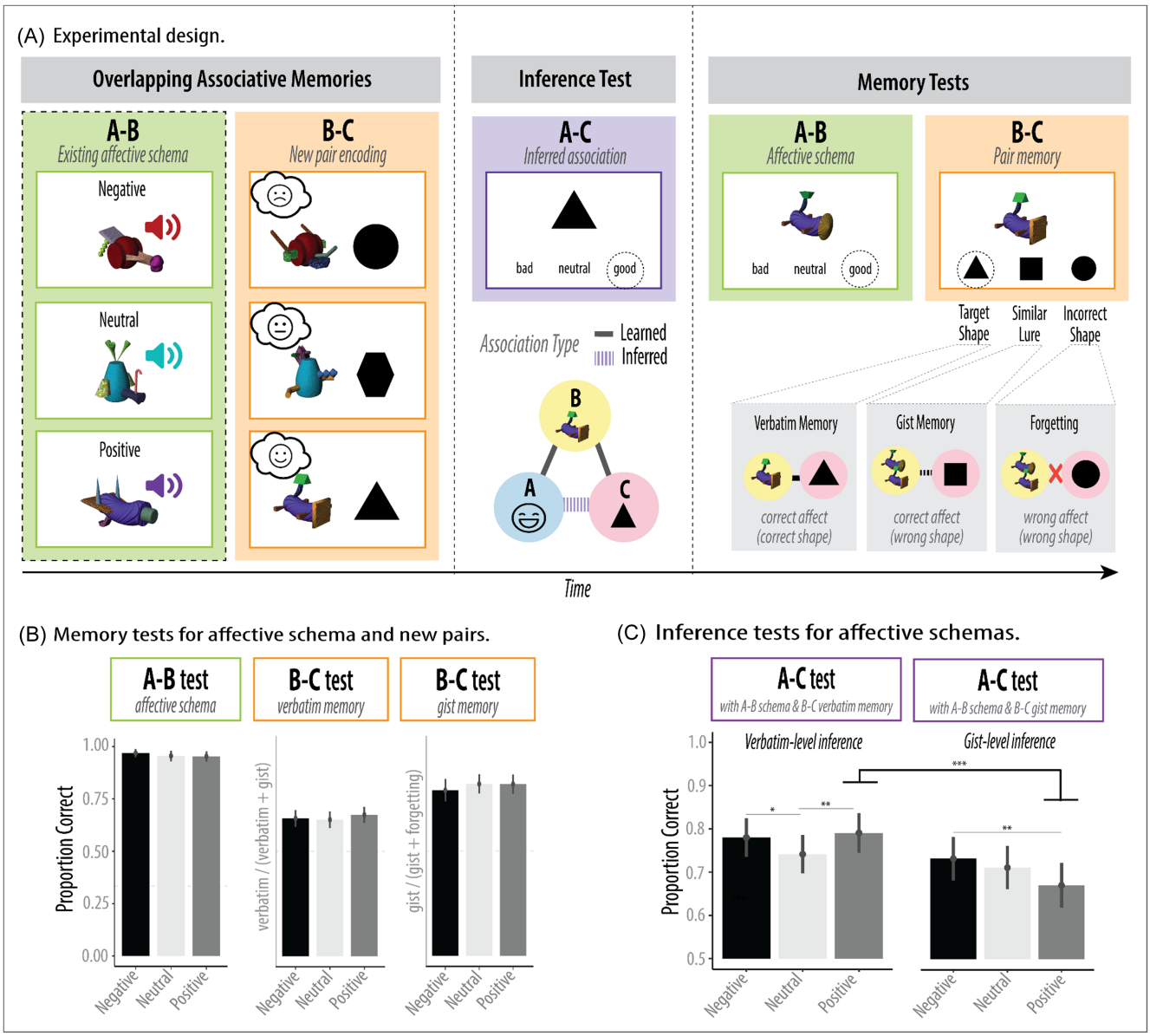
Figure 4
Inference From Affective Schemas



Note. Results of Bayesian multilevel models for the associative inference task. Points are posterior means and error bars are 95% credibility intervals. The strength of the evidence for the alternative hypothesis (H1) over the null hypothesis (H0) is indicated by probability of direction estimates (p_{dir} values). A. *Left*: The inference task involved an ABC triad (noises + Fribbles + shapes) of overlapping A–B and B–C associative memories. A–B associations were the affective schemas acquired at Visit 1 (A–B = noises + Fribbles). At Visit 2, the A–B affective schemas were restudied and then new B–C pairs (Fribbles + shapes) were passively encoded. *Middle:* The inference test asked participants to infer the affective association of the shapes (never-before-experienced A–C pairs). *Right:* A–B and B–C memory tests were administered after the A–C inference test. The B–C pair memory test included three response options: (a) the correct target shape that was an exact match for the encoded B–C pair, which represented detailed verbatim memory; (b) a similar lure shape that was paired with a Fribble of the same affective valence, which represented gist memory; and (c) an incorrect shape of a different affective valence, which represented forgetting. B. Associative memory was above chance levels, but A–B or B–C memory did not differ as a function of affective schemas. C. Verbatim- and gist-level inference accuracy was above chance for all conditions. Negative schemas were associated with relatively strong inference performance both at the verbatim and gist levels, whereas inference performance for positive schemas varied by verbatim and gist levels. *Left*: Verbatim-level inferences were correct A–C responses when the associated A–B and B–C verbatim responses were correct. Participants were more accurate at making negative and positive verbatim-level inferences relative to neutral inferences. *Right*: Gist-level inferences were correct A–C responses when the associated A–B memory was correct and B–C gist responses were endorsed. Participants were more accurate when making negative gist-level inferences than positive gist-level inferences. Icons licensed from Freepik (https://www.freepik.com/). See the online article for the color version of this figure.
* p_{dir} = .95–.96. ** p_{dir} = .97–.98. *** p_{dir} = .99–1.00.

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valence that characterized each of the 36 shapes (good, bad, or neutral). The inferential associations (A–C) were not explained to the participant and no feedback was provided. Correct inferences could be based on having correct (a) verbatim memory or (b) gist memory for the B–C fribble-shape pairs.

B–C Test (Pair Memory)

Participants then completed a three-alternative, forced-choice memory task in which they were asked to select which of the three shapes was paired with a given Fribble. The response choices always included the (a) correct shape (*verbatim/detailed memory*), (b) incorrect shape that was paired with a Fribble of the same affective schema as the correct shape (*gist/generalized memory*), and (c) incorrect shape that was paired with a Fribble of a different affective schema.

A–B Test (Affective Schemas)

Participants completed a final affective schema knowledge test with 36 Fribbles using the same procedures as the Visit 1 test phase (see above). Evidence for affective schema consolidation was examined by comparing within-person change between the post-acquisition test and this A–B test.

Fribble Affect Semantic Ratings

To assess whether the affective semantic components of the schemas were consolidated and/or maintained, participants completed Fribble affect ratings both before and after the inference task, using the same procedures from Visit 1.

Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. We preregistered our hypotheses and analysis plan prior to viewing or analyzing any data (https://osf.io/qtdwb). At the time of preregistration, a portion of the data (the in-person sample) was already collected but was not examined in any way. Additional participants were subsequently collected online using the same experimental tasks and procedures. All data were analyzed only after preregistration. Although we preregistered to recruit 240 online participants to power secondary individual-difference analyses, financial constraints limited enrollment of an online sample to 126. This approach still allowed us to exceed the sample size needed to detect our primary within-subject effects ($N = 110$), but as a result, we omitted the planned individual-difference analyses from the present study. In addition, post hoc data inspection revealed that cumulative accuracy across trials followed a nonlinear, sigmoidal trajectory in both the acquisition and reversal learning tasks. Consequently, we supplemented our preregistered plan with sigmoidal-parameter estimation to better characterize affective schema learning dynamics.

All data cleaning and analyses were conducted using R-4.3.3 (R Core Team, 2021) except for the sigmoid parameter estimation (Supplemental Table S3 details the computing environment). We followed the Bayesian Analysis Reporting Guidelines (Kruschke, 2021) to facilitate transparency and reproducibility. All data, analysis code, and materials that support the findings in this study are publicly available on the Open Science Framework repository for this project (Vannucci et al., 2025; https://osf.io/7a5jk/).

Analysis Plan

Sample Size Determination

Parametric Monte Carlo power simulations (1,000 runs at each sample size, $\alpha = .05$) were conducted with the simr package in R (Green & MacLeod, 2016) on mixed-effects multilevel models from pilot data ($N = 14$) for three outcomes, including (a) the number of trials to criterion (Task 1: acquisition), schema-inconsistent accuracy (Task 2: reversal learning), and A–C verbatim inference accuracy (Task 3: associative inference). Across sample sizes ($N =$ 5–125, in steps of 10), 1,000 data sets were simulated by refitting the model and testing the within-subject fixed effects of the affective schema condition. To achieve >80% power to detect these effects, a sample size of 60 was required for Task 1, 90 for Task 2, and 110 for Task 3; accordingly, we targeted 110 participants for the present study.

Data Cleaning

Data quality was cleaned in line with recommendations for online data collection (Kominsky et al., 2021; Nussenbaum et al., 2020). The Qualtrics survey included three attention check items, which no participants failed. Trials with response times of less than 150 ms were removed from analysis because this threshold approximates the physiological lower limit for a simple detection response to a visual stimulus (Berger & Kiefer, 2021; Luce, 1986; Teichner & Krebs, 1972). Consistent with this premise, responses below the 150 ms cutoff were at chance-level accuracy across all tasks. As such, low effort and lack of attention was defined a priori as a task having greater than 59% of trials with response times less than 150 ms *and* poor task performance (at chance-level overall). This screening led to the exclusion of four participants from the acquisition task and three participants from the inference task (Supplemental Figure S9). In line with the common practice and recommendations (Berger & Kiefer, 2021; Kominsky et al., 2021; Nussenbaum et al., 2020), individual trials were also excluded when response times were <150 ms and when response times were three standard deviations above or below the participants' mean response time. Simulation studies reveal that this trial-level exclusion approach introduces minimal biases, whereas a lack of outlier exclusion shows the largest bias in patterns of statistical significance and Type I errors (Berger & Kiefer, 2021). Participants were also excluded from analysis if they did not learn the affective schemas and stopped the task early at Visit 1 ($n = 1$) or did not accurately identify the correct B–C pairs from the associative inference task above chance levels at Visit 2 ($n = 9$; Supplemental Figure S9). No participants were excluded from the updating task based on these screening criteria. After this data quality screening, the final sample size for each task was 176 for the acquisition task, 108 for the updating task, and 93 for the inference task.

Sigmoid Parameter Estimation

The estimation of sigmoidal function parameters was not planned in the preregistered analyses. Qualitative inspection of the acquisition and updating task data suggested that the way in which cumulative

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accuracy changed as a function of time (i.e., trial-by-trial) was mostly nonlinear. As such, separate sigmoid functions were fit for each condition within each individual for both tasks using SciPy-1.12.0 (Virtanen et al., 2020) and curve fitting functions in Python-3.11.8 (Python Software Foundation, 2024). The sigmoid function is a generalized form of the standard logistic function to include parameters that adjust for its amplitude, midpoint, steepness, and horizontal shift (Hosmer et al., 2013), which were formalized as:

$$y = \frac{L}{1 + e^{-k(x-x0)}} + b, \quad (1)$$

where y refers to cumulative accuracy; x to time (trial number); L to amplitude (or upper asymptote); k to steepness of exponential growth; $x0$ to the midpoint (or inflection point); and b to the horizontal shift (Supplemental Figure S1). Originally applied to understand bioecological phenomena (Malthus, 2018), the sigmoid function has been used frequently in computational models of semantic knowledge acquisition, category learning, and memory transformation (e.g., the integration of new information into existing knowledge structures; Ashby & Maddox, 2011; Courville et al., 2004; Kruschke, 1991; Saxe et al., 2019).

Sigmoid curve parameters can be interpreted within the context of behavioral learning experiments (Murre, 2014). The upper asymptote (L) is the curve's maximal achievable value and suggests a point beyond which no significant improvements or learning are expected given the current level of experience. Here, the sigmoid curve's midpoint ($x0$) indicates the trial number at which the curve reaches its inflection point, which is the point at which learners start to rapidly grasp or master the conceptual knowledge. Higher values of the inflection point indicate a relative delay in the rapid learning phase. The steepness (k) of the curve can be construed as the learning rate or how quickly learning progresses over time, with higher scores indicating faster learning. Finally, the horizontal shift parameter (b) is also referred to as the "lag phase" and provides insight into the timing and onset of learning progress. Positive b values shift the curve to the right, indicating a longer initial lag phase when little learning progress was observed compared to curves without this shift. On the other hand, negative b values shift the curve to the left, indicating an earlier onset of learning progress and may occur in situations where learners have some prior knowledge of the content.

Data Analysis

Rationale

A Bayesian statistical framework was used to examine the role of affective semantic features in schema learning processes (acquisition, inference, and updating). Currently, there is a lack of prior knowledge about affective schema mechanisms and therefore this study developed novel experimental paradigms. As such, we were primarily interested in characterizing the presence and direction of within-person affective schema effects on schema task outcomes. Key benefits of Bayesian models include: (a) high flexibility in handling complex data structures such as hierarchical, multilevel models; (b) a complete distribution of parameter estimates (i.e., posterior distribution) that allows for a direct, intuitive probabilistic interpretation of results (e.g., "there is a 95% probability that the parameter lies within this interval"); (c) ability to incorporate prior knowledge through the use of prior distributions; and (iv) robustness with smaller samples and noisy data (Dienes & Mclatchie, 2018; Kruschke & Liddell, 2018).

Model Variables and Priors

Bayesian mixed-effects multilevel models were estimated with a Gaussian process using the brms package (Bürkner, 2017). All models had the same predictor variables. Repeated trials and affective schema conditions (Level 1) were nested within a person (Level 2). Continuous predictor variables were grand mean-centered and categorical variables were dummy coded. The affective schema conditions were the independent variables of interest. Covariates included age, gender (0.5 = female vs. −0.5 = nonfemale), subjective social status, and experiment modality (0.5 = online vs. −0.5 = laboratory). Primary dependent variables involved measures related to accuracy and sigmoid learning curve parameters. Median response times for correct trials were also examined as a secondary outcome. The affective schema conditions and task performance outcomes varied slightly by task (Tables 1–3).

Noninformative priors were specified given the lack of prior knowledge and research with these novel affective schema paradigms. Bayesian models combined this prior information with the evidence provided by the data using the likelihood function to calculate a probability distribution over each parameter's potential values (i.e., posterior distribution; Gelman et al., 2013). Parameter estimates, therefore, were quantified as the probability that the true (unknown) effect lies between a range of different values in a random population sample, given the observed data (Gelman et al., 2013). These distributions of possible values for the model parameters are called posterior distributions, which were extracted using the tidybayes package (Kay & Mastny, 2023).

Model Fit Diagnostics

Model fit and convergence were evaluated in line with recent recommendations (Schad et al., 2021). Markov chain Monte Carlo (MCMC; Diaconis, 2009) is a computational tool for generating posterior distributions (i.e., parameter estimates) in Bayesian models. Four MCMC chains were run independently to evaluate the convergence of the model parameter estimates. The MCMC simulations generated between 10,000 and 50,000 iterations (i.e., samples) per chain (iterations were increased until all parameters for each model converged). Model convergence was examined (a) visually, using trace plots of all MCMC draws generated by the bayesplot package (Gabry et al., 2024), and (b) quantitatively, by calculating the potential scale reduction factor (R^) and bulk effective sample size (ESS) for all model parameters (Vehtari et al., 2021). R^ values reflect the ratio of within-chain and between-chain variability, with values closer to 1 suggesting that the MCMC algorithm sufficiently converged (Vehtari et al., 2021). The bulk ESS provides insight into the quality and efficiency of the MCMC sampling process, with higher ESS values indicating more reliable and stable estimates of the posterior distribution. R^ values less than 1.01 and bulk-ESS values greater than 1,000 suggested that model parameter estimates generated by the four MCMC chains converged well and were considered reliable (Schad et al., 2021). Supplemental Table S4 provides details about each model's MCMC setup and indicators of model fit for primary affective schema contrasts of interest (bulk-ESS; R^).

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Table 1
Within-Person Affective Schema Effects for Acquisition

	Affective schema contrast					
	Negative (vs. neutral)		Positive (vs. neutral)		Negative (vs. positive)	
Variable	B_{mdn} [90% HDI]	% in ROPE	B_{mdn} [90% HDI]	% in ROPE	B_{mdn} [90% HDI]	% in ROPE
Trials needed to reach criterion (no.)	−0.87** [−1.48, −0.25]	0.00	−0.70* [−1.35, −0.06]	1.31	−0.16 [−0.81, 0.48]	20.53
Response time (ms)	−93.80*** [−156.64, −30.37]	0.00	−65.58* [−126.68, −6.45]	6.19	−28.13 [−91.55, 35.10]	34.63
Sigmoidal learning curve parameters						
Learning onset (*b*)	−0.002 [−0.011, 0.008]	97.14	−0.011* [−0.021, −0.001]	42.65	0.009* [0.001, 0.018]	54.68
Inflection point (*x*0)	−0.91*** [−1.55, −0.29]	0.00	−0.43 [−1.00, 0.15]	11.83	−0.48 [−1.10, 0.13]	10.69
Learning rate (*k*)	−0.0003 [−0.013, 0.012]	100.00	0.005 [−0.011, 0.020]	100.00	−0.005 [−0.020, 0.010]	100.00
Asymptote (*L*)	0.015 [−0.005, 0.035]	31.01	0.010 [−0.009, 0.029]	50.97	0.005 [−0.012, 0.023]	65.50
Affective semantic ratings						
Slope (pre- → posttask)	−1.70*** [−1.88, −1.52]	0.00	1.58*** [1.40, 1.76]	0.00	−3.28*** [−3.47, −3.10]	0.00
Affective schema knowledge test						
Accuracy (proportion correct)	0.013 [−0.008, 0.034]	39.16	−0.001 [−0.019, 0.016]	73.11	0.014 [−0.000, 0.029]	29.75

Note. Results of Bayesian multilevel models with mixed effects, adjusting for participant age, gender identity, subjective social status, and modality (online vs. laboratory). Median response times were computed for correct trials. B_{Mdn} represents the median of the posterior distribution for each parameter. In Bayesian inference, 90% high density credibility intervals (HDIs) are reported over standard 95% credibility intervals because the tails of the posterior distributions (areas near the edges of the distribution) are less reliably estimated than the center due to fewer samples in these regions. The probability of direction (p_{dir}) is the proportion of the posterior distribution that the parameter's effect is in the specified direction (positive or negative). The region of practical equivalence (ROPE) is a threshold that distinguishes values that may be practically meaningful. The percentage (%) of values within the ROPE reflects the proportion of the full posterior distribution that lies within the ROPE. The ROPE range encompassed the range of values considered to be negligible or too small to be meaningful and were: (a) [−0.01, 0.01] for proportion scores (i.e., <1% difference between conditions); (b) [−20, 20] for response time outcomes (i.e., <20 ms difference between conditions); and (c) [−0.1, 0.1] for all other measures following generalized recommendations (Makowski, Ben-Shachar, Chen, & Lüdecke, 2019). vs. = versus.
* p_{dir} = .95–.96. ** p_{dir} = .97–.98. *** p_{dir} = .99–1.00.

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Table 2
Within-Person Schema Comparison of Affective Schema Updating

	Assimilation: Affective schemas consistent			Accommodation: Affective schema reversal		
	Negative (neg) New (vs. old) consistent stimuli	Neutral (neu) New (vs. old) consistent stimuli	Positive (pos) New (vs. old) consistent stimuli	Valence reversal Neg → Pos (vs. Pos → Neg)	Neutral- to-valenced Neu → Neg (vs. Neu → Pos)	Valenced- to-neutral Neg → Neu (vs. Pos → Neu)
Variable	B_{mdn} [90% HDI]	B_{mdn} [90% HDI]	B_{mdn} [90% HDI]	B_{mdn} [90% HDI]	B_{mdn} [90% HDI]	B_{mdn} [90% HDI]
Proportion correct	−0.008 [−0.016, 0.003]	−0.003 [−0.011, 0.006]	−0.004 [−0.012, 0.004]	−0.006 [−0.049, 0.037]	0.0002 [−0.033, 0.034]	−0.006 [−0.048, 0.036]
Response time (ms)	−24.62 [−61.61, 12.61]	22.02 [−14.97, 58.55]	21.93 [−14.97, 58.55]	35.97 [−87.69, 161.21]	−249.11*** [−468.95, −33.20]	112.69 [−43.04, 269.68]
Proportion perseverative error	—	—	—	0.024 [−0.009, 0.055]	0.004 [−0.023. 0.031]	0.044** [0.009, 0.079]
Sigmoidal learning curve parameters						
Learning onset (b)	−0.006 [−0.011, 0.0002]	−0.007** [−0.013, −0.002]	−0.007** [−0.012, −0.001]	0.001 [−0.019, 0.020]	−0.013 [−0.029, 0.003]	0.001 [−0.020, 0.021]
Inflection point ($x0$)	5.39*** [5.19, 5.60]	5.25*** [5.05, 5.46]	5.25*** [5.05, 5.45]	−0.33 [−1.01, 0.34]	−0.36 [−1.12, 0.41]	−0.59 [−2.09, 0.86]
Learning rate (k)	−0.080*** [−0.082, −0.077]	−0.082*** [−0.084, −0.079]	−0.081*** [−0.084, −0.079]	0.62 [−0.56, 1.77]	0.47 [−0.77, 1.74]	−0.60 [−1.61, 0.42]
Asymptote (L)	−0.005*** [−0.009, −0.002]	−0.001 [−0.005, 0.003]	0.0001 [−0.003, 0.004]	−0.018 [−0.077, 0.041]	0.015 [−0.034, 0.063]	0.007 [−0.049, 0.063]
Affective semantic ratings						
Absolute difference (pre- → posttask)	−0.09 [−0.25, 0.06]	−0.002 [−0.14, 0.14]	0.05 [−0.09, 0.20]	−0.90*** [−1.26, −0.55]	0.28* [0.02, 0.53]	−0.17 [−0.51, 0.17]

Note. Results of Bayesian multilevel models with mixed effects, adjusting for participant age, gender identity, subjective social status, and modality (online vs. laboratory). Separate models were conducted for the assimilation and accommodation components of the updating task. Median response times were computed for correct trials. B_{Mdn} represents the median of the posterior distribution for each parameter. In Bayesian inference, 90% high density credibility intervals (HDIs) are reported over standard 95% credibility intervals because the tails of the posterior distributions (areas near the edges of the distribution) are less reliably estimated than the center due to fewer samples in these regions. The probability of direction (p_{dir}) is the proportion of the posterior distribution that the parameter's effect is in the specified direction (positive or negative). — = no perseverative errors were possible for affective schema assimilation because the valence remained the same; vs. = versus.
* p_{dir} = .95–.96. ** p_{dir} = .97–.98. *** p_{dir} = .99–1.00.

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Table 3
Within-Person Affective Schema Comparison of Associative Inference

	Affective schema contrast					
	Negative (vs. neutral)		Positive (vs. neutral)		Negative (vs. positive)	
Variable	B_{mdn} [90% HDI]	% in ROPE	B_{mdn} [90% HDI]	% in ROPE	B_{mdn} [90% HDI]	% in ROPE
A–B memory test: Affective semantic memory						
Proportion correct	0.014 [−0.004, 0.031]	35.24	−0.002 [−0.020, 0.015]	72.46	0.016 [−0.004, 0.035]	30.37
Response time (ms)	22.56 [−8.57, 54.25]	43.99	5.54 [−21.52, 32.68]	83.68	17.28 [−13.56, 48.09]	56.51
B–C memory test: Pair memory						
Verbatim memory (correct affect, correct shape)						
Proportion correct (hits/hits + gist)	0.006 [−0.027, 0.040]	40.12	0.024 [−0.012, 0.059]	22.68	−0.017 [−0.051, 0.017]	30.48
Response time (ms)	−503.88*** [−808.60, −198.96]	0.00	−291.24 [−650.41, 68.88]	3.30	−214.66 [−490.36, 59.61]	4.59
Gist memory (correct affect, wrong shape)						
Proportion correct (gist/gist + misses)	−0.030 [−0.079, 0.019]	18.06	−0.001 [−0.040, 0.039]	35.79	−0.028 [−0.076, 0.020]	18.57
Response time (ms)	−451.24 [−1206.16, 274.88]	2.34	−837.62 [−1798.01, 138.22]	1.08	303.50 [−174.31, 801.55]	3.44
A–C inference test: Affective semantic inference						
Verbatim-level inference (with A–B correct + B–C verbatim memory)						
Proportion correct	0.038* [0.001, 0.076]	5.79	0.050** [0.013, 0.086]	0.00	−0.013 [−0.049, 0.027]	31.78
Response time (ms)	−210.82** [−375.82, −43.04]	0.00	−288.94*** [−444.54, −121.14]	0.00	76.09 [−66.56, 216.68]	13.90
Gist-level inference (with A–B correct + B–C gist memory)						
Proportion correct	0.021 [−0.027, 0.067]	23.86	−0.051 [−0.087, 0.005]	9.14	0.062** [0.014, 0.110]	0.00
Response time (ms)	−43.22 [−365.13, 259.60]	9.29	46.48 [−307.26, 397.21]	8.20	−122.74 [−462.10, 217.57]	7.27
Verbatim (vs. gist) inference interaction						
Proportion correct	−0.017 [−0.073, 0.038]	22.72	−0.090*** [−0.146, −0.034]	0.00	0.073** [0.017, 0.127]	0.00
Response time (ms)	125.78 [−208.56, 461.99]	7.14	361.15* [24.90, 697.19]	0.00	−253.01 [−582.82, 78.41]	2.18

Note. Results of Bayesian multilevel models with mixed effects, adjusting for participant age, gender identity, subjective social status, and modality (online vs. laboratory). Median response times were computed for correct trials. B_{Mdn} represents the median of the posterior distribution for each parameter. In Bayesian inference, 90% high density credibility intervals (HDIs) are reported over standard 95% credibility intervals because the tails of the posterior distributions (areas near the edges of the distribution) are less reliably estimated than the center due to fewer samples in these regions. The probability of direction (p_{dir}) is the proportion of the posterior distribution that the parameter's effect is in the specified direction (positive or negative). The region of practical equivalence (ROPE) is a threshold that distinguishes values that may be practically meaningful; The percentage (%) of values within the ROPE reflect the proportion of the full posterior distribution that lies within the ROPE. The *ROPE* range encompassed the range of values considered to be negligible or too small to be meaningful and were: (a) [−0.01, 0.01] for proportion scores (i.e., <1% difference between conditions); (b) [−20, 20] for response time outcomes (i.e., <20 ms difference between conditions); and (c) [−0.1, 0.1] for all other measures following generalized recommendations (Makowski, Ben-Shachar, Chen, & Lüdecke, 2019). A–B = affective schemas (affective noises + Fribble families); B–C = pair memory (Fribbles + shapes); A–C = affective inference (never-before-seen affective valence/noises + shapes); vs. = versus.
* p_{dir} = .95–.96. ** p_{dir} = .97–.98. *** p_{dir} = .99–1.00.

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Hypothesis Testing

The most salient difference between Frequentist and Bayesian statistics is the interpretation of uncertainty and probability (Fornacon-Wood et al., 2022). The Frequentist approach assigns probability to data, whereas the Bayesian approach assigns probability to hypotheses (based on the data and prior knowledge). Frequentist statistics involve Null Hypothesis Significance Testing where the data are used to calculate *p* values that test whether the probability of observing data is at least as extreme as the data observed, assuming the null hypothesis is true (Wasserstein & Lazar, 2016). That is, the Frequentist *p* value provides a measure of the data's compatibility with the conditions of the null hypothesis; it does not measure the probability that the null hypothesis is true or false nor does it does not provide information about the credibility of the alternative hypothesis (Wasserstein & Lazar, 2016). With a Bayesian approach, by contrast, probabilistic statements *can* be made about support for the alternative *and* null hypotheses. Distinctions have been drawn between effect *existence* and effect *significance* in Bayesian hypothesis testing (Makowski, Ben-Shachar, Chen, & Lüdecke, 2019). Existence refers to the objective probability of the presence of an effect in a particular direction (i.e., negative or positive) without making assumptions about its magnitude or importance, whereas significance is re-framed as an effect's importance in terms of exceeding some given threshold (e.g., practical relevance; Makowski, Ben-Shachar, Chen, & Lüdecke, 2019).

In Bayesian statistics, model effects are described by characterizing the posterior distributions (Schad et al., 2021; van de Schoot et al., 2021). Inspection of posterior predictive checks generated using the bayesplot package (Gabry et al., 2024) showed that all models approximated the data well. The bayestestR package (Makowski, Ben-Shachar, & Lüdecke, 2019) calculated indices of effect *existence* and *significance* from the posterior distributions of the within-person affective schema parameters. Following recommendations for Bayesian models with noninformative priors and little prior knowledge (Kelter, 2023; Makowski, Ben-Shachar, Chen, & Lüdecke, 2019), we report the following parameters: the posterior median (*Mdn*) to quantify the central tendency, 90% highest density credibility intervals (HDIs) to quantify the uncertainty of the parameter estimate, the probability of direction (p_{dir}) to quantify the likelihood of effect existence, and the region of practical equivalence ($ROPE_{full}$) to quantify effect significance. The posterior median and 90% HDIs were reported (vs. mean and 95% HDIs) because the tails of posterior distributions tend to be unstable, such that 90% HDIs were demonstrated to be more reliable and reproducible than 95% HDIs (Gong & Flegal, 2016; Makowski, Ben-Shachar, & Lüdecke, 2019). Effect existence and direction (primary study goal) was determined with p_{dir} values, which are the proportion of the posterior distribution that is the same sign (positive or negative) as the median's; p_{dir} values >.95 indicated the existence of a directional effect (Makowski, Ben-Shachar, Chen, & Lüdecke, 2019). Effect significance was determined by the $ROPE_{full}$, which is the proportion of the full posterior distribution that lies within the *ROPE* (Makowski, Ben-Shachar, Chen, & Lüdecke, 2019; Schad et al., 2021). The ROPE range encompassed the range of values considered to be negligible or too small to be meaningful and were: (a) [−0.01, 0.01] for proportion scores (i.e., <1% difference between conditions); (b) [−20, 20] for response time outcomes (i.e., <20 ms difference between conditions); and (c) [−0.1, 0.1] for all other measures following generalized recommendations (Kelter, 2023; Makowski, Ben-Shachar, Chen, & Lüdecke, 2019; Schwaferts & Augustin, 2020). Evidence for the alternative hypothesis will be accepted for $ROPE_{full}$ values <0.05, whereas the null hypothesis was accepted for $ROPE_{full}$ values >0.95 (Makowski, Ben-Shachar, Chen, & Lüdecke, 2019).

Post Hoc Exploratory Analysis

Sensitivity analyses indicated that neither study modality nor Fribble body color accounted for the observed affective schema effects. The direction and significance of effects were comparable when including versus excluding trials with outlier response times (i.e., values more extreme than $M \pm 3\ SD$).The findings were robust to the type of Bayesian prior specified (noninformative vs. weakly informative), suggesting that the choice to use a noninformative prior in primary analyses did not bias the results. Exploratory analyses also probed individual differences in affective schema processes, which found that earlier/faster valenced (vs. neutral) schema acquisition was associated with (a) greater difficulty with accommodating neutral information into the valenced affective schemas (Supplemental Figure S7) and (b) better inference ability with valenced (vs. neutral) schemas (Supplemental Figure S8). Detailed results are in the online Supplemental Materials.

Results

Bayesian multilevel models evaluated the within-person effects of affective schema conditions on indices of acquisition, updating, and inference. All model parameters converged based on standard guidelines for Bayesian approaches (Supplemental Table S4; Schad et al., 2021).

Task 1: Forming Affective Schemas

Table 1 presents the model results for the affective schema learning task (Figure 2A).

Schema Acquisition Dynamics

Negative (M = 14.3, 95% CI [13.1, 15.6]) and positive schemas (M = 14.9, 95% CI [13.6, 16.7]) resulted in more rapid acquisition (M = 15.6, 95% CI [14.1, 16.4]; Figure 2B, left) and faster correct response times than the neutral schema (Supplemental Figure S3A). No differences were observed between the negative and positive schemas for either variable ($p_{dir}s$ = 0.66–0.77). Sigmoid learning curves (Supplemental Figure S1) fit to cumulative accuracy scores showed an excellent fit (Figure 2B; $M_{Rsquared}$ = 0.989; 95% CI [0.988, 0.990]). The positive schema had an earlier learning onset than the negative and neutral schemas, which did not differ from each other (Table 1; Supplemental Figure S3C). By contrast, the negative schema had an earlier inflection point than the neutral schema (Supplemental Figure S3C), indicating that participants reached the stage of rapidly understanding or mastering the negative schema sooner than the neutral schema. No inflection point differences were found between the positive schema and either the neutral or the negative schema. There were no differences in learning peak ($p_{dir}s$ = 0.69–0.90) or learning rate ($p_{dir}s$ = 0.52–0.70; Supplemental Figure S3C) among the schemas.

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Acquisition Tests

Affective schema knowledge was assessed through (a) affective ratings of Fribble stimuli before and after acquisition and (b) a valence knowledge test (Figure 2B). Participants achieved near-perfect accuracy on the affective schema knowledge test (*M*s = 0.95–0.96; Supplemental Figure S3B), indicating successful schema acquisition and generalization to similar exemplars. Affective ratings shifted as expected (Figure 2C). Initially (pretask), Fribbles were rated as affectively neutral, but posttask ratings decreased to "very negative" for the negative schema ($M = 2.25$, 95% CI [2.13, 2.37]) and increased to "very positive" for the positive schema ($M = 5.53$, 95%CI [5.40, 5.65]), with no change for the neutral schema ($M = 4.08$, 95%CI [3.97, 4.19]).

Consolidation Tests

Affective schema consolidation was examined via the extent of within-person changes in (a) affective ratings of Fribbles (Figure 2B) and (b) schema knowledge test accuracy from the end of Visit 1 (i.e., after schema acquisition) to the beginning of Visit 2 (i.e., ∼24 hr after schema acquisition but before Visit 2 experimental tasks). The differentiated affective ratings for each schema condition at the end of Visit 1 (as noted above, "very negative" for the negative schema; "very positive" for the positive schema; "neutral" for the neutral schema) remained stable at Visit 2 after ∼24-hr (Figure 2C; $p_{\text{dir}}s$ = 0.63–0.76). Participants achieved near-perfect accuracy on the affective schema knowledge tests at both Visit 1 (*M*s = 0.95–0.96; Supplemental Figure S3B) and Visit 2 (*M*s = 0.95–0.97; Figure 4B, left panel), with no meaningful within-person differences between visits ($p_{\text{dir}}s$ = 0.72–0.92). These findings suggest that the affective components of the schemas persisted and were maintained in memory ∼24 hr following initial acquisition, providing evidence for affective schema consolidation.

Task 2: Updating Affective Schemas

Key comparisons of interest for the schema updating task (Figure 3A) are in Table 2, with additional results in the online Supplemental Table S8. Sigmoid curve parameters (Supplemental Figure S1) were fit to cumulative accuracy scores, showing an excellent fit ($M_{\text{Rsquared}} = 0.985$; 95% CI [0.982, 0.987]).

Assimilation of Schema-Consistent Experiences

Assimilation of schema-consistent experiences was examined by comparing (a) old and new schema-consistent stimuli and (b) new schema-consistent and -inconsistent stimuli. Learning was near-perfect for old and new stimuli across all schemas (*M*s = 0.995–0.997; Supplemental Figure S5A), suggesting robust assimilation of similar experiences into prior knowledge (Figure 3B). New, schema-consistent stimuli were learned with slower learning rates and later inflection points than old, previously seen stimuli across schemas (Figure 3B; Supplemental Figure S5A). Despite earlier initial learning onset for new (vs. old) stimuli, these differences were small-to-negligible and the parameters did not differ from zero (Supplemental Figure S5A). Response times did not differ between old and new stimuli across schemas (Supplemental Figure S5B). Whereas schema-consistent stimuli were learned with higher accuracy ($p_{\text{dir}}s$ = 0.99–1.00) and with quicker response times than inconsistent stimuli, ($p_{\text{dir}}s$ = 0.99–1.00), the onset of learning occurred earlier for schema-inconsistent stimuli than for consistent stimuli ($p_{\text{dir}}s$ = 0.99–1.00).

Accommodation of Schema-Inconsistent Experiences

Accommodation of schema-inconsistent experiences varied by affective content. Valenced-to-neutral reversals (negative/positive-to-neutral) were the most resistant to accommodation, as evidenced by the lowest overall accuracy Figure 3C; Supplemental Figure S6D; Neg → Neu: $M = 0.71$, 95% CI [0.64, 0.78]; Pos → Neu: $M = 0.70$, 95% CI [0.64, 0.76]). By contrast, valenced reversals (negative-to-positive and positive-to-negative) showed the highest accuracy (Figure 3C; Supplemental Figure S6D; Neg → Pos: $M = 0.81$, 95% CI [0.76, 0.88]; Pos → Neg: $M = 0.84$, 95% CI [0.78, 0.89]), with intermediate accuracy scores for neutral-to-valenced reversals (neutral-to-negative/positive; Figure 3C; Supplemental Figure S6D; Neu → Neg: $M = 0.77$, 95% CI [0.72, 0.84]; Neu → Pos: $M = 0.76$, 95% CI [0.70, 0.83]). There were no differences in other sigmoid parameters (Supplemental Figure S6). Valenced-to-neutral reversals had slower response times than valenced reversals (Supplemental Figure S6E). Perseverative errors endorsed the valence of the initial schema, indicating resistance to accommodation, and were more likely in valenced-to-neutral reversals, particularly when the original schema was negative (Figure 3D). No biases in perseverative errors were observed for valenced or neutral-to-valenced schema reversals (Figure 3D).

Accommodation Test

Affective schema accommodations were tested by examining changes in affective ratings of Fribble exemplars from pre- to posttask. All changes occurred in expected directions (Supplemental Figure S6F). For example, in the positive-to-negative reversal, ratings decreased from "very positive" to "very negative." Greater absolute changes in affective ratings (reflecting magnitude) were observed when accommodating negative information into a positive schema (Pos → Neg: $M_{\text{abs-diff}} = 3.51$, 95% CI [3.09, 3.94]) compared to the reverse (Neg → Pos: $M_{\text{abs-diff}} = 2.61$, 95% CI [2.14, 3.07]). Greater absolute changes also occurred when accommodating negative information into a neutral schema (Neu → Neg: $M_{\text{abs-diff}} = 1.81$, 95% CI [1.51, 2.10]) than positive information (Neu → Pos: $M_{\text{abs-diff}} = 1.53$, 95% CI [1.23, 1.82]). No differences were found when accommodating negative or positive information into a neutral schema (Neg → Neu: $M_{\text{abs-diff}} = 1.97$, 95% CI [1.62, 2.34]; Pos → Neu: $M_{\text{abs-diff}} = 2.14$, 95% CI [1.78, 2.50]).

Task 3: Inferences From Affective Schemas

Model results for the affective schema-based associative inference task are shown in Table 3.

A–B Test: Affective Schema Knowledge

A–B test performance was near-perfect for all affective schema conditions (*M*s = 0.95–0.97), with no differences as a function of affect (Figure 4B). There were also no differences for A–B response times ($p_{\text{dir}}s$ = 0.58–0.91; Supplemental Figure S4A).

B–C Test: Pair Memory

The alternative forced-choice task assessed whether affective schemas support rapid learning of new Fribble-shape pairs. We examined (a) whether these associations could be retrieved after two-shot encoding and (b) whether memory was detailed (verbatim memory; hits for the specific shape) or generalized (gist memory that was schema-consistent but blurred with limited details; Robin & Moscovitch, 2017). Each fribble test probe offered three shape options: the correct shape, a schema-consistent lure (shape paired with a different Fribble from the same affective schema family), and a schema-inconsistent shape (i.e., forgetting; Figure 4A, right). *Verbatim memory* (verbatim/verbatim + gist responses) was above chance across all affective schemas (negative: $M = 0.66$, 95% CI [0.62, 0.70]; neutral: $M = 0.65$, 95% CI [0.61, 0.69]; positive: $M = 0.67$, 95% CI [0.64, 0.71]), with no differences between them (Figure 4B). Participants responded faster when recognizing B–C pairs within the negative schema than the neutral schema (Supplemental Figure S4A), with marginal differences from the positive schema ($p_{\text{dir}}s = 0.90$). These results suggest that negative schemas improved accessibility to rapidly formed, verbatim memory. *Gist memory* (gist/gist + forgetting responses) was also above chance (Figure 4B), with no differences in accuracy (negative: $M = 0.79$, 95% CI [0.74, 0.84]; neutral: $M = 0.82$, 95% CI [0.77, 0.86]; positive: $M = 0.82$, 95% CI [0.77, 0.86]) or response times (Supplemental Figure S4A) across schemas. *Forgetting* showed no affective valence biases ($p_{\text{dir}}s = 0.61$–0.91; Supplemental Table S5).

A–C Inference Test

The B–C pair memory test allowed us to examine inferences based on verbatim and gist memory (Figure 4A). That is, A–C inferences were examined only when the corresponding A–B association was remembered correctly and when participants remembered the Fribble as being paired with correct target shapes (B–C verbatim memory) or with critical lure shapes (B–C gist memory), but not incorrect shapes. Whereas positive inferences were only strong for verbatim memories, negative inferences were strong for both verbatim and gist memories. Relative to the neutral schema ($M = 0.74$, 95% CI [0.70, 0.79]), negative and positive schemas (negative: $M = 0.78$, 95% CI [0.74, 0.82]; positive: $M = 0.79$, 95% CI [0.74, 0.84]) resulted in more accurate verbatim-level inferences (Figure 4C), and these valenced inferences were made faster (Supplemental Figure S4B). Errors in verbatim-level inferences for valenced schemas were biased toward selecting neutral responses (Supplemental Table S6 and Figure S4C). Even without access to verbatim B–C memory, *gist-level inferences* were above chance (Figure 4C, right), with higher accuracy for negative ($M = 0.73$, 95% CI [0.68, 0.78]) than positive inferences ($M = 0.67$, 95% CI [0.61, 0.72]). Accuracy for neutral gist-level inferences ($M = 0.71$, 95% CI [0.66, 0.76]) did not differ from either valenced schema. Gist-level inferences were slower than verbatim-level inferences (Supplemental Figure S4B), with no affective schema differences in response times (Supplemental Figure S4B, right). No affective biases were found in gist-level inference errors (Supplemental Table S7 and Figure S4D). A–C inferences could not be made when B–C pairs were forgotten.

Discussion

It is essential to clarify higher order, conceptual-level emotional memory processes to the same depth that complex cognitive knowledge structures have been explored. Bridging this gap will advance our understanding of how emotional experiences shape behavior and cognition at higher levels of abstraction, with significant implications for affective science, cognitive science, emotional development, and mental health. Here, we show that emotional learning too can organize into complex schematized memory from specific, unique affective experiences and semanticize into higher order schematic memory. Specifically, we found that a class of Fribble stimuli (Barry et al., 2014), for which participants would have no prior associations, could acquire a common affective value from a distribution of unique, similarly valenced noises (standardized affective auditory stimuli; Yang et al., 2018). These Fribble–valence associations were not merely stimulus–stimulus associations; they were complex, higher order generalized knowledge structures abstracted from numerous exemplars, consolidated over a 24 hr period, that facilitated new learning and generated affective inferences. We used novel tasks that blend classic elements from widely used conditioning (Dunsmoor et al., 2012; Izquierdo et al., 2017; Keller et al., 2020) and schema (Brod et al., 2015, 2017; Spalding et al., 2018) paradigms to advance theories of human cognition and behavior by illustrating when affective schema processes align with or diverge from established knowledge on affective learning and semantic memory. This work offers new insights into the structure and function of complex emotional memories and how they shape our responses to emotional situations.

Valenced Affective Schemas Are Acquired Faster Than a Neutral Schema

During initial schema acquisition, valenced (negative/positive) schemas were learned to criterion (>90% accuracy) in fewer trials than the neutral schema. Learning curve parameters showed that positive schemas had an earlier onset of *any* initial learning than negative and neutral schemas (albeit by a very small amount), whereas the negative schema had an earlier inflection point. This latter finding indicates that key concepts of the negative schema were grasped more quickly (Murre, 2014). Consistent with decades of research in the episodic memory domain (Dolcos et al., 2017; LaBar & Cabeza, 2006; Phelps, 2004; Rouhani et al., 2023), these findings suggest that affective information, particularly negative, is prioritized for initial schema learning. The affective schemas differed from the neutral schema in both arousal and valence, so it is not possible to determine whether the observed learning differences are driven by arousal, valence, or their interaction—each known to rely on distinct neural mechanisms (Bowen et al., 2018; Kensinger & Corkin, 2004; Kensinger & Ford, 2021). We attempted to minimize this confound by matching negative and positive sound stimuli on normative arousal ratings (Yang et al., 2018) and through pilot testing. However, subtle variations in arousal may nonetheless have contributed to differential affective schema acquisition.

Acknowledging this limitation, several mechanisms may help explain these findings. One possibility is that arousal-related engagement of the locus coeruleus–noradrenergic system (including

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the amygdala) draws attention toward emotionally salient stimuli and enhances their encoding, giving valenced affective schemas an early advantage in learning and memory formation (Clewett & Murty, 2019; Talmi, 2013). At the same time, valence may shape how affective schemas begin to form by influencing the way emotional experiences are interpreted as they unfold. Indeed, negative and positive information is known to engage distinct neurocognitive pathways and meaning-making processes during encoding (Kensinger & Ford, 2021; Talmi, 2013). Emotional stimuli could also be represented as being more similar to one another, both psychologically and neurally, than neutral events are (Riberto et al., 2019, 2022), which may help promote integration and the early organization of affective schema-like representations. Taken together, we speculate that enhanced acquisition of affective schemas, particularly for negatively valenced information, in this study may potentially reflect contributions from both arousal and valence. Future studies will be important to disentangle these dimensions using a fully crossed design (e.g., High/Low Arousal × Positive/Negative Valence) and physiological or neural measures to test these mechanisms directly.

Affective Schemas Assimilate and Accommodate to New Information, but Valenced-to-Neutral Schema Reversals Are Most Resistant to Accommodation

Affective schema updating results paralleled prior work on schema assimilation by showing enhanced, rapid learning of new schema-consistent information (Brod et al., 2017; Sommer, 2017; van Kesteren et al., 2013), but they contrasted our hypotheses about schema accommodation (that valenced schema reversals would be the most resistant). Strikingly, incorporating new affectively neutral information into consolidated valenced schemas was the most difficult. These data underscore the affective nature of the brain when it comes to knowledge accumulation. Once conceptual information acquires emotional significance, schemas may resist dropping the affective content. This resistance may be especially pronounced in negative schemas, as perseverative errors were more common in negative schemas than positive. Given that schema updating occurred after a ∼24-hr delay, it is possible that arousal-related engagement of the locus coeruleus–noradrenergic system—possibly via interactions between the amygdala, hippocampus, and medial prefrontal cortex—prioritized the consolidation, abstraction, and/or semanticization of the most salient aspects of emotional experiences into long-term memory (Clewett & Murty, 2019; Krenz et al., 2021; LaBar & Phelps, 1998). Because negative information carries heightened biological relevance for one's wellbeing, such neural mechanisms could potentially bias the brain toward a more conservative "better-safe-than-sorry" strategy when updating negative schemas (Van den Bergh et al., 2021), helping to avoid underestimating future threats. Like extinction processes (Dunsmoor et al., 2015; Lipp et al., 2020), neutral associations may also be encoded as new, but weak representations that compete for retrieval. This finding highlights a tendency to preserve semanticized negative information, underscoring the role of negative schemas in shaping adaptive responses to potentially aversive outcomes. Consistent with this interpretation, we also found that shifts to negative schemas elicited greater absolute affective changes than shifts to positive ones, suggesting that negative content may more powerfully engage stability-promoting mnemonic mechanisms that prioritize the retention and integration of threat-relevant information.

Valenced (Negative↔Positive) Schema Reversals Are Most Flexible to Accommodation

Valenced schema reversals, specifically shifts between negative and positive schemas, were surprisingly adaptable, with the highest accuracy, fastest response times, and most significant changes in affective ratings. These results suggest that affective schemas may possess an inherent flexibility in processing and integrating emotionally charged information. This finding notably diverges from the counterconditioning literature (Keller et al., 2020), which finds that aversive-to-appetitive and appetitive-to-aversive reversals are more difficult to learn than initial conditioning with neutral and affective stimuli. The results also differ from the valence dimension of widely used emotion frameworks that situate affective states on a spectrum ranging from negative to positive at opposite poles, with neutral being in the middle (Russell, 1980). In other words, positive and negative emotions are traditionally conceptualized as being more similar to neutral affective states than to each other within the valence domain. However, the current findings suggest the opposite—that affective schemas are more easily reconfigured when reversing between negative and positive information.

Rather than suggesting inconsistency with these frameworks, these findings may highlight the critical role of arousal in shaping affective schema flexibility for accommodating emotionally inconsistent information. Arousal is traditionally considered orthogonal to valence in models of emotion (e.g., Russell, 1980), pointing to a potential mechanism by which affective schemas may readily adapt despite opposing valence. Specifically, the shared high arousal of negative and positive schemas (compared to neutral) may facilitate reversals between them, in contrast to the resistance observed when updating to neutral schemas. Although speculative, this interpretation aligns with the idea that overlapping components of arousal abstracted from negative and positive experiences serve as common "affective ingredients" spanning both negative and positive schemas, perhaps rendering them more semantically and neurally similar representations than previously assumed. If so, from a constructionist perspective on how emotions "are made" in the brain (Barrett, 2017; Lindquist, 2013), negative and positive schemas may differ more in their semantic labeling than in their underlying affective core. Thus, while valenced schema reversals may be aided by shared arousal, neutral updates may lack the salience needed to accommodate consolidated affective knowledge. Supporting this view, prior work on reappraisal finds that it is easier to change the valence of high-arousal emotional states than to downregulate arousal itself (McRae, 2016; McRae et al., 2012), suggesting that arousal-rich schema representations may be especially amenable to affective reinterpretation.

Gist-Based Affective Inferences Are Possible Without Access to Verbatim Representations

Findings from the associative inference task (B–C test specifically) demonstrate that affective schemas enabled rapid learning of new related experiences that can be accessed at both detailed (verbatim) and generalized (gist) levels. These findings replicate prior studies showing that new experiences create memory traces

with varying specificity (Robin & Moscovitch, 2017). By design, the related experiences (Fribble + shape "B–C" pairs) partially overlapped with existing affective schemas (affect + Fribble "A–B" associations), allowing for novel "A–C" inferences about the affective valence of shapes. Similar to prior studies (Schlichting et al., 2017; Spalding et al., 2018; van Kesteren et al., 2020; Wimmer & Shohamy, 2012), verbatim-level inferences (made when A–B and B–C pairs are correctly recognized verbatim) were highly accurate across all schemas. Moving beyond prior work, verbatim-level inferences with valenced schemas are more accurate than with a neutral schema. There are at least two potential routes to successful inferences. First, participants may retrieve a direct A–C link from an integrated memory representation, which formed during the encoding of the new related experience (Schlichting & Preston, 2015) or during rapid memory consolidation (Audrain & McAndrews, 2022). It is possible that affective memories promote memory integration (Zhu et al., 2022), which in turn enables more accurate inferences with valenced schemas. Second, A–B and B–C experiences could be stored separately as nonoverlapping, distinct traces, with participants only logically combining information from these two memories at retrieval to make A–C inferences (Banino et al., 2016). Based on this latter view, superior inference with valenced schemas stem from stronger A–B and B–C memory traces that were more accessible when needed at retrieval than neutral ones. Further work is needed to disentangle these possibilities.

In a substantial extension of the prior literature, the current results indicate that affective A–C inferences can be made accurately from gist-level representations, even without access to detailed verbatim (B–C) representations. Note that inference performance was at chance when B–C pairs were completely forgotten, highlighting the role of gist representations in enabling novel inferences. The ability to make gist-level affective inferences could be an important feature of affective schemas because they are, by definition, already flexible associative structures that foster rapid assimilation of related information (Ghosh & Gilboa, 2014; Moscovitch & Gilboa, 2024). Despite comparable access to B–C gist representations across all schemas in this study, negative schemas were more effective than positive ones in supporting gist-level inferences. In other words, strong negative inferences are made based on both gist and verbatim memories, whereas memory specificity benefits inferences supported by positive schemas. This valence difference was surprising given that the broaden-and-build theory proposes that positive emotions facilitate flexible and integrative processing (Clewett & Murty, 2019; Fredrickson, 2004), which could be expected to enhance gist-based inferences for positive schemas. One possibility is that preexisting negative semantic knowledge structures are more elaborated and interconnected than positive ones, which may more readily facilitate schema-related memory integration and gist-based inferences. Indeed, false memory studies involving words or pictures tend to find that negative valence, but not positive valence, strengthens gist traces in semantic processing contexts (Bookbinder & Brainerd, 2016, 2017). Should valence differences in access to gist traces extend to inferences, the results would suggest that negative schemas may adaptively prioritize neural integration at encoding or favor a cautious inference strategy for unfamiliar, potentially threatening information.

Constraints on Generality

The present findings should be interpreted with consideration of the sample characteristics and study design. The sample comprised primarily young adults (ages 18–36), which limits the generalizability of results to older or younger populations. The racial and ethnic composition of the sample included a diverse range of identities (e.g., 20% Asian or Asian American, 14% Hispanic/Latine, 11% Black or African American) but did not include representation from Indigenous populations. Participants' self-appraisals of their position in the United States with regard to objective socioeconomic indicators suggest that the sample represents the full socioeconomic spectrum (Supplemental Table S1). Participants were recruited through two distinct pathways following disruption of in-person research related to the COVID-19 pandemic: undergraduates from a large urban private university in the northeast (laboratory sample) and young adults from the United States via Prolific (online sample). The inclusion of both in-person and online samples enhanced the broader diversity of the sample, but differences in recruitment and testing environments could introduce variability in participants' experiences. However, the sample modality did not account for any of the study results in sensitivity analyses, indicating that the findings generalize to both contexts. Future research should assess whether the observed effects generalize to broader age ranges and other cultural or geographic contexts.

Although novel Fribble exemplars were presented across all experiments, Fribbles within each affective schema shared some perceptual commonalities (e.g., body color) and were linked explicitly to verbal valence concepts (bad/good/neutral). Participants could have relied on perceptual commonalities or verbal emotion labels rather than affective similarities for schema learning. However, it is unlikely that systematic differences in affective schema learning and inference processes would be observed at the within-person level if either of these strategies were driving the findings. Perceptual salience and complexity were matched across Fribble families, and the assignment of affective schema condition to different Fribble families was randomly counterbalanced across participants to minimize perceptual confounds. In post hoc sensitivity analyses, the inclusion of Fribble body color (the most salient perceptual similarity) as a within-subjects condition did not alter any affective schema effects. Furthermore, the affective schema paradigms replicated classic effects from the conditioning paradigms from which the methods were drawn, increasing confidence in our conclusions that the current results reflect semanticized affective processing.

There is also evidence to suggest that the noises imbued Fribble families with affective significance. The auditory stimuli were drawn from an international standardized database of noises shown to elicit affective responses (Yang et al., 2018); the affective and arousal features of the noises were confirmed in a separate pilot study prior to conducting the experiments. Valence and arousal are often conflated in emotional memory studies (Bennion et al., 2013), so the positive and negative noises were matched on arousal to minimize this potential confound. Correct responses were also made faster with negative and positive schemas than with the neutral schema across all three experiments, a phenomenon that is routinely observed during affective associative learning and memory tasks. Affective ratings of Fribble exemplars also changed in expected ways during the schema acquisition and updating tasks. From pre- to

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postacquisition, for example, ratings increased from "neutral" to "very positive" in the positive schema condition, "neutral" to "very negative" for the negative schema, and had no changes from "neutral" for the neutral schema. Reliance on verbal emotion labels (vs. affective experience) could account for changes in subjective affective ratings, but cannot fully account for within-person accuracy and response time differences. Furthermore, the use of emotion labels does not necessarily facilitate or alter learning new affective concepts (Hoemann et al., 2022). Future work would benefit from experiments that dissociate verbal emotion labels from contexts that naturally induce affective responses during schema learning. Moving forward, the use of functional magnetic resonance imaging will be crucial for understanding neural mechanisms.

Conclusions

Schemas are widely recognized as knowledge structures that organize conceptual-level information, adapt to new experiences, and facilitate novel inferences in unfamiliar situations (Gilboa & Marlatte, 2017). Our results reveal that incorporating affective features into schema learning significantly extends these principles. Specific learned affective associations become semanticized into affective schemas (negative, positive, and neutral). Once acquired, they can assimilate congruent information, accommodate to new-inconsistent emotional information, and generate new affective inferences. Valenced (negative/positive) affective schemas are acquired faster than neutral schemas, are more difficult to reverse, and facilitate rapid learning and memory for related information. Negative-valenced schemas in particular are prioritized for learning, more resistant to change, and more conducive to gist-based inferences. These findings break new ground by revealing mechanisms of affective semantic memory—from initial schema formation to their adaptive functions in novel emotional situations. By demonstrating the unique properties of affective schemas, this work establishes a novel methodological and theoretical framework for conceptualizing how complex emotional memory operates.

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